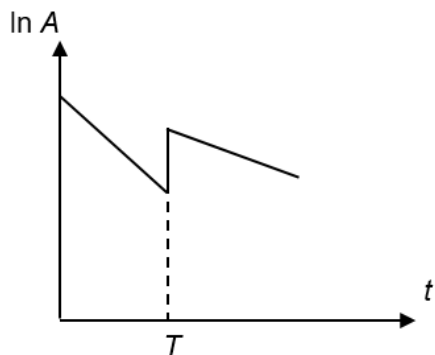


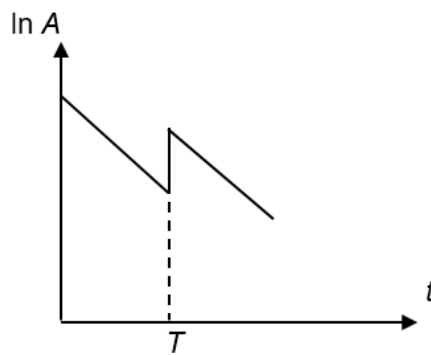
30 At time $t = 0$, some radioactive gas is injected into a sealed vessel. At time T , a different radioactive gas with a half-life very much shorter than the first is injected into the same vessel.

Which one of the following graphs best represents how activity A varies with t ?

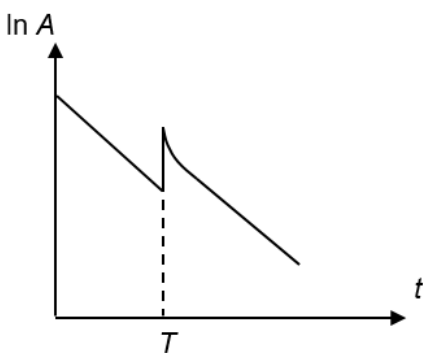
A



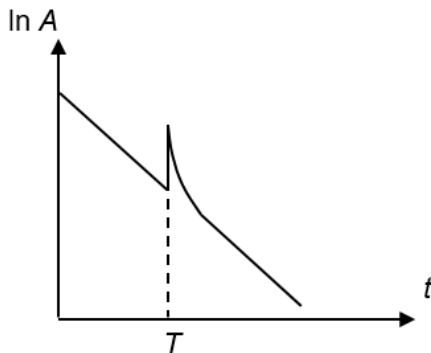
B



C



D

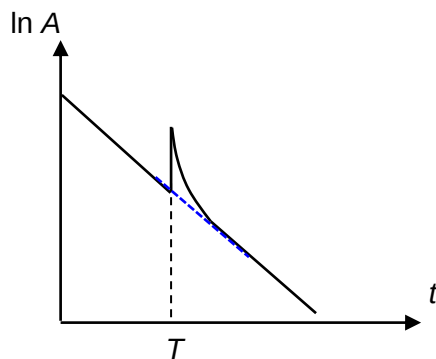


Answer: D

$$A = A_0 e^{-\lambda t}$$

$$\ln A = \ln A_0 - \lambda t$$

The curved part comes about because of two different decay constants. Since the added gas has a much shorter half-life, the gradient of the graph should eventually return to the original gradient, and “continue” from where the original “left off”.



-- END OF PAPER 1 --

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